THE DERIVED CATEGORY OF TORSION ABELIAN SHEAVES IS THE FULL SUBCATEGORY OF COMPLEXES WITH TORSION COHOMOLOGY

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: divisibility, the quasi-isomorphism statement, the K-injectivity and the adjunction identities are all written out; the closing 'Some details omitted' concerns one identification and is accompanied by the author's reference to a more general result that supplies it, and it is preserved verbatim in the counted span.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Torsion abelian sheaves form a Serre subcategory. Same chapter, the torsion abelian sheaves on a site, which are closed under subobjects, quotients and extensions. Supplies the subcategory A whose derived category is described.
- The derived adjunction of the inclusion of the full subcategory. Derived Categories, Lemma derived-lemma-derived-adjoint-functors, together with the criterion for adjoint functors with invertible unit and counit. Supplies the equivalence criterion applied at the end.

Source reference. Original theorem 079P: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4260–4260. Navigation pointer invoked at proof line 4260; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 09T5: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4274–4274. Navigation pointer invoked at proof

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line 4274; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 07RB: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 4277–4277. Navigation pointer invoked at proof line 4277; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $\mathcal{A} \subset Ab(\mathcal{C})$ denote the Serre subcategory consisting of torsion abelian sheaves. Then the functor $D(\mathcal{A}) \to D_{\mathcal{A}}(\mathcal{C})$ is an equivalence.*

Proof. A key observation is that an injective abelian sheaf $\mathcal{I}$ is divisible. Namely, if $s \in \mathcal{I}(U)$ is a local section, then we interpret s as a map $s : j_{U!}\mathbf{Z} \to \mathcal{I}$ and we apply the defining property of an injective object to the injective map of sheaves $n : j_{U!}\mathbf{Z} \to j_{U!}\mathbf{Z}$ to see that there exists an $s' \in \mathcal{I}(U)$ with $ns' = s$.

For a sheaf $\mathcal{F}$ denote $\mathcal{F}_{tor}$ its torsion subsheaf. We claim that if $\mathcal{I}^\bullet$ is a complex of injective abelian sheaves whose cohomology sheaves are torsion, then

$$\mathcal{I}^\bullet_{tor} \to \mathcal{I}^\bullet$$

is a quasi-isomorphism. Namely, by flatness of $\mathbf{Q}$ over $\mathbf{Z}$ we have

$$H^p(\mathcal{I}^\bullet) \otimes_{\mathbf{Z}} \mathbf{Q} = H^p(\mathcal{I}^\bullet \otimes_{\mathbf{Z}} \mathbf{Q})$$

which is zero because the cohomology sheaves are torsion. By divisibility (shown above) we see that $\mathcal{I}^\bullet \to \mathcal{I}^\bullet \otimes_{\mathbf{Z}} \mathbf{Q}$ is surjective with kernel $\mathcal{I}^\bullet_{tor}$. The claim follows from the long exact sequence of cohomology sheaves associated to the short exact sequence you get.

To prove the lemma we will construct right adjoint $T : D(\mathcal{C}) \to D(\mathcal{A})$. Namely, given K in $D(\mathcal{C})$ we can represent K by a K-injective complex $\mathcal{I}^\bullet$ whose cohomology sheaves are injective, see Injectives, Theorem 079P. Then we set $T(K) = \mathcal{I}^\bullet_{tor}$, in other words, T is the right derived functor of taking torsion. The functor T is a right adjoint to $i : D(\mathcal{A}) \to D_{\mathcal{A}}(\mathcal{C})$. This readily follows from the observation that if $\mathcal{F}^\bullet$ is a complex of torsion sheaves, then

$$\mathrm{Hom}_{K(\mathcal{A})}(\mathcal{F}^\bullet, I^\bullet_{tor}) = \mathrm{Hom}_{K(Ab(\mathcal{C}))}(\mathcal{F}^\bullet, I^\bullet)$$

in particular $\mathcal{I}^\bullet_{tor}$ is a K-injective complex of $\mathcal{A}$. Some details omitted; in case of doubt, it also follows from the more general Derived Categories, Lemma 09T5. Our claim above gives that $L = T(i(L))$ for L in $D(\mathcal{A})$ and $i(T(K)) = K$ if K is in $D_{\mathcal{A}}(\mathcal{C})$. Using Categories, Lemma 07RB the result follows. □